



VLSI2026\_01\_20\_v1(27).pdf  
PDF

tell what is the minimum width and length of tft, refer this paper

Thought for 3m 24s >

The paper **does not explicitly list the lateral TFT width  $W$  and channel length  $L$**  as numerical design parameters.

However, using the **20  $\mu\text{m}$  scale bar in Fig. 3(b)**, the fabricated transistor geometry can be approximately estimated as:

- **Minimum channel length,  $L_{\min}$ : approximately 5–6  $\mu\text{m}$**
- **Channel width,  $W$ : approximately 20–24  $\mu\text{m}$**

Therefore, for a paper-consistent simulation/layout value, use:

$$W = 24 \mu\text{m}, \quad L = 5 \mu\text{m}$$

Also, the **7 nm ZnO value mentioned in the fabrication section is the channel-layer thickness**, not the channel length. The paper states a 7 nm ZnO layer for both the access ZnO TFT and Fe-ZnO TFT fabrication.  VLSI2026\_01\_20\_v1

So your present  $L = 5 \mu\text{m}$ ,  $W = 24 \mu\text{m}$  TFT is the closest defensible choice from this paper. A width of 6  $\mu\text{m}$  cannot be claimed as the TFT width specified by the paper.

       Sources

```
// VerilogA for igzo, tft_n1, veriloga
```

```
include "constants.vams"
include "disciplines.vams"
```

```
module tft_n1(d, s, g);
```

inout d, s, g;  
electrical d, s, g;

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Thought for 13s >

Compare the devices using their **aspect ratio**  $W/L$ , because drain current or channel conductance approximately scales with  $W/L$ .

### Reference TFT used for the table model

$$\left(\frac{W}{L}\right)_{\text{TFT}} = \frac{8}{3} = 2.667$$

Your layouts give:

| Designed NFET                   | $W/L$ | Relative to the $8/3$ TFT |
|---------------------------------|-------|---------------------------|
| $W = 6 \mu m, L = 5 \mu m$      | 1.20  | $0.45\times$              |
| $W = 20.825 \mu m, L = 5 \mu m$ | 4.165 | $1.562\times$             |
| $W = 24 \mu m, L = 5 \mu m$     | 4.80  | $1.80\times$              |

Therefore, among your **single-transistor layouts**, the closest is:

$$W = 6 \mu m, \quad L = 5 \mu m$$

However, it is only slightly closer than the  $20.825 \mu m$  device:

- $6/5$  is approximately **55% weaker** than the modeled  $8/3$  TFT.
- $20.825/5$  is approximately **56.2% stronger**.
- $24/5$  is approximately **80% stronger**.

### Width required for an exact geometric match

To preserve the original  $W/L = 8/3$  while changing the length to  $5 \mu m$ :

$$W_{\text{equivalent}} = 5 \left(\frac{8}{3}\right) = 13.33 \mu m$$

Thus, the ideal equivalent geometry would be:

$$W \approx 13.33 \mu m, \quad L = 5 \mu m$$

Since you only have widths 6, 20.825, and 24  $\mu m$ , using two 6  $\mu m$  TFTs in parallel is substantially closer:

$$W_{\text{effective}} = 2(6) = 12 \mu m$$

$$\frac{W_{\text{effective}}}{L} = \frac{12}{5} = 2.4$$

This is only about 10% **lower** than the original 8/3 TFT ratio. Therefore:

$$\text{Best single device: } 6 \mu m / 5 \mu m$$

$$\text{Best practical equivalent: two } 6 \mu m / 5 \mu m \text{ devices in parallel}$$

One important point: your Verilog-A model does not contain  $w$  or  $L$  parameters. The conductance table already represents the 8  $\mu m$  / 3  $\mu m$  device, so changing the schematic or layout width will **not automatically scale the simulated current**. For a 6/5 device, the table conductance should approximately be scaled by:

<> verilog

```
gch = $table_model(vgs, "g_n1_tbl.tbl", "1LL") * 0.45;
```



For two 6/5 devices in parallel, the corresponding approximate factor is:

<> verilog

```
gch = $table_model(vgs, "g_n1_tbl.tbl", "1LL") * 0.90;
```



This scaling assumes conductance is proportional to  $W/L$ ; actual TFT characteristics can also differ because of contact resistance, mobility, threshold voltage, and fabrication variation.

